

GeM-Rakshak

Automated Statutory Verification, Document Forensics & Cartel Detection Middleware for Government e-Marketplace (GeM)

< 30 sec

Turnaround Time

Down from 4.5 days manual scrutiny

8 Gateways

Live Statutory Checks

GSTN, MCA21, CBDT, Udyam, CPPP, EPFO

100% CVC

Audit Sealed

Cryptographic SHA-256 non-repudiation

Aligned Entity: Chennai Petroleum Corporation Limited (CPCL) - MoP&NG

Live Presentation Platform URL: <http://localhost:3000> - Repository: github.com/SiddhantGpt19/GeM-Rakshak

The Public Procurement Integrity Crisis

Why manual bid evaluation fails and how government entities are compromised

DELAYS

Severe Processing Delays

- Procurement officers manually inspect 28+ statutory documents per tender bid.
- Average technical evaluation takes 4.5 to 7 days per tender.
- Delays lead to stalled refinery shutdowns, project overruns, and supply chain freezes.

FORGERY

Document Tampering & Forgery

- Bidders alter turnover digits using Adobe Photoshop to meet qualification thresholds.
- Forged Chartered Accountant UDIN numbers and fake seals.
- Unrelated QR codes pasted onto fraudulent certificates to deceive human checkers.

COLLUSION

Hidden Cartel Syndicates

- Collusive bidders rotate bids using proxy companies sharing common directors.
- Same IP subnets, identical bank branches, and synchronized bid upload timestamps.
- Debarred / blacklisted vendors bidding through newly created front entities.

GeM-Rakshak 3-Layer Defense Framework

Seamless middleware sitting between GeM and Official Source-of-Truth Registries

LAYER 1 - Zero Trust

Document Forensics Lab

Pixel & Metadata Tamper Detection

- Error Level Analysis (ELA) thermal heatmaps for altered numbers
- EXIF & XMP metadata provenance checking (Catches Photoshop / InDesign)
- Dual-engine QR code payload vs visible OCR text cross-reconciliation
- ICAI UDIN algorithmic cryptographic checksum validation

LAYER 2 - 8 Gateways

Statutory Registry Gateways

Real-Time Source-of-Truth Cross-Check

- GSTN Gateway: Live GST status, 3B regularity & declared turnover
- MCA21 V3: Corporate active status, paid-up capital & Director DINs
- CBDT / Income Tax: Section 206AB higher TDS defaulter screening
- Ministry of MSME: Udyam Manufacturing vs Services scope check
- CPPP Central Registry: CPSE blacklisting & debarment lookup

LAYER 3 - AI Network

Cartel Radar & CVC Audit

Syndicate Detection & Non-Repudiation

- Graph Network Analysis detecting shared directors, IPs & banks
- Collusion confidence scoring & automated 4-point judicial breakdown
- Immutable SHA-256 cryptographically chained CVC audit log
- One-click CVC Audit Dossier PDF generator with digital hashes

Executive Command Console (/dashboard)

The central nerve center providing instant situational awareness to procurement officers

Console Capabilities & Real-Time Telemetry

1. Active Entity Switcher

Allows CPSE officers (e.g. CPCL Manali Refinery, NTPC, IOCL) to switch active procurement sessions with 1 click.

2. 4 Elevated Real-Time KPI Cards

Active Tenders (3), Bids Screened Today (4), Fraud Blocked (7 Forgeries Flagged), and Turnaround Time (<30s down from 4.5 days).

3. Compliance Status Distribution Donut

Instant visual breakdown of bids: 25% Compliant, 25% Clarification Needed, 50% High Risk / Disqualified.

4. Time Saved Comparative Bar Chart

Direct visual proof: Manual evaluation takes ~108 hours vs GeM-Rakshak taking ~0.5 hours.

5. Live Anomaly Feed

Real-time ticker flagging statutory violations (e.g., Photoshop alteration detected on Bidder 003; Cartel connection between Bidder 003 & 004).

[*] JUDGE EVALUATION GUIDE

What to Notice on Dashboard:

- The top banner directly displays the procuring entity and tender reference.
- Notice the 'Cartel Radar: 1 Ring Detected' banner with a direct quick-link to inspect the syndicate.
- Instant English / Hindi / Marathi language toggle without page reload.
- Clicking 'Scrutinize Bids' moves immediately to the tender comparison pipeline.

Bidder Scrutiny & Comparison Pipeline

Screening all bidders against mandatory tender parameters simultaneously

BID-001 - 98.0%

Aura Flow Systems

Turnover: Rs. 18.5 Cr (Req: Rs. 15 Cr) [OK]

MII: Class-I Local (68.5%) [OK]

GST: 100% Tax Compliant [OK]

Udyam: Mfg (NIC 28131) [OK]

CPPP: Not Debarred [OK]

100% genuine bidder. All 8 registry cross-checks match submitted documentation.

BID-002 - 48.5%

Bharat Petro-Tech

Turnover: Rs. 16.0 Cr (Req: Rs. 15 Cr) [OK]

MII: Class-I Claimed (55.0%)

GST: 2 Filings Pending [!]

Udyam: Services (NIC 74909) [!]

CPPP: Not Debarred [OK]

Udyam registered for Services instead of Mfg; 2 GST filings pending; OEM ref generic.

BID-003 - 8.0%

Apex Engineering

Turnover: Rs. 1.5 Cr (Forged: Rs. 18.5 Cr) [X]

MII: Fails Class-I (22.0%) [X]

GST: GST Suspended [X]

Udyam: Udyam Cancelled [X]

CPPP: Debarred until Oct 2027 [X]

CRITICAL FRAUD: Actively debarred by IOCL; Photoshop-forged CA certificate; forged UDIN.

BID-004 - 25.0%

Apex Flow Dynamics

Turnover: Rs. 1.52 Cr [X]

MII: Fails Class-I (35.0%) [X]

GST: Active (Low compliance)

Udyam: Trading (NIC 46599) [X]

CPPP: Common Director DIN [X]

CARTEL PROXY: Shares Partner Vikram Singhal with debarred Bidder 003. Common IP subnet.

Interactive Bidder Audit Workspace (/bidders/[id]/audit)

50/50 split-screen environment giving procurement officers complete audit control

LEFT PANEL: Document Forensics Canvas

- **Document Tabs**

Switch between Udyam, GST REG-06, CA Turnover Certificate, OEM Form, and EPFO Challans.

- **Clean Scanline Overlay**

Subtle, realistic document inspection scanline without glowing sci-fi laser tropes.

- **Interactive OCR Text Bounding Boxes**

Hover over extracted fields to inspect OCR confidence score and statutory field mapping.

- **Thermal ELA Heatmap View**

One-click toggle revealing image compression differences where text was altered in Photoshop.

- **Embedded QR Code Scanner**

Real scannable QR code verifiable with any mobile phone camera in real time.

RIGHT PANEL: 5 Accordions & Officer Actions

- **1. MCA21 & Director Identity**

Corporate status, paid-up capital & CPPP blacklisting check.

- **2. MSME & Udyam Scope**

Validates manufacturing classification (NIC 28132) vs services.

- **3. GSTN & Tax Solvency**

Checks GSTIN active status, GSTR-3B filings & Sec 206AB TDS.

- **4. Make in India (MII) Audit**

Verifies CA Local Content percentage ($\geq 50\%$ for Class-I).

- **5. EPFO / ESIC Manpower**

Checks active contributing employees & latest wage month challan.

- **Sticky Officer Action Bar**

One-click actions: Draft Clarification Notice, Disqualify Bidder, Qualify Bidder, or Export CVC Audit PDF.

Document Forensics Lab (/forensics-lab)

Real-time PDF uploading and deep forensic inspection exposing fraudulent bid certificates

01

Error Level Analysis (ELA)

Detecting Pasted Pixel Layers

- Calculates JPEG/DCT compression differences across 8x8 image blocks.
- Tampered text (e.g. altering Rs. 1.5 Cr to Rs. 18.5 Cr) has different error rates than original document.
- Renders an instant visual thermal heatmap highlighting pasted seals and numbers.

02

EXIF & XMP Metadata Audit

Authoring Software Forensics

- Extracts hidden binary PDF Producer and Creator Tool tags.
- Official government certificates are generated via automated wkhtmltopdf or Apache FOP.
- Instantly flags documents processed in Adobe Photoshop CC 2024 or desktop publishing tools.

03

QR vs OCR Cross-Check

Exposing Copied / Spoofed QR

- Scans encoded QR matrix payload using computer vision.
- Compares payload fields against OCR text visible on the document face.
- Exposes bidders who copied a legitimate QR code from an old Rs. 1.5 Lakh invoice onto an Rs. 18.5 Cr tender.

04

ICAI UDIN Verifier

Chartered Accountant Check

- Every official CA certificate in India requires an 18-digit Unique Document Identification Number.
- GeM-Rakshak validates UDIN format, CA membership number, and issuance date.
- Flags forged UDINs and non-existent membership records instantly.

Cartel Collusion Radar (/cartel-network)

Exposing bid-rigging syndicates and proxy bidding rings under Section 3(3) of Competition Act 2002

The 4-Point Syndicate Fingerprint (Bidder 003 & 004)

[DIN] Common Director (DIN 08492019)

Partner Vikram Singhal holds 50% shares in actively debarred Bidder 003 and also controls proxy Bidder 004.

[LOC] Common Physical Location

Both entities registered at B-44, Okhla Industrial Area Phase-II, New Delhi, sharing administrative offices.

[BANK] Common Banking Branch (HDFC0001245)

Both bidders maintain statutory bank accounts at the same HDFC branch, indicating pooled financial backing.

[IP] Identical File Creator & Subnet IP

Bids submitted from the same 192.168.44.0/24 subnet; PDFs generated using the same pirated Photoshop install.

[!] CARTEL DETECTION IMPACT

Why Human Officers Miss Cartels:

- Bidders 003 and 004 appear as separate legal entities with distinct PAN and GSTIN numbers.
- A human officer reviewing bids individually has no way of knowing they share an IP or director.
- GeM-Rakshak performs cross-bidder relational graph mapping automatically.
- Prevents debarred vendors from laundering contracts through newly registered proxy front companies.

8 Official Government Gateways (/gateways)

Real-time query console querying verified national registries with sub-50ms latency

GSTN Gateway (CBIC / GSTN) GSTIN operational status, GSTR-1 & 3B return regularity, declared turnover vs bid claim.	MCA21 V3 Registry (Ministry of Corporate Affairs) Corporate active status, paid-up capital, incorporation date, Director DIN linkage.
Income Tax / CBDT (Department of Revenue) PAN validity, Section 206AB higher TDS rate defaulter screening.	Udyam MSME Portal (Ministry of MSME) Enterprise category (Micro/Small), Manufacturing vs Services NIC 5-digit code.
CPPP Central Registry (Ministry of Finance / CPSEs) All-India debarment & blacklisting history under Rule 151 of GFR 2017.	Shram Suvidha (EPFO) (Ministry of Labour) Active EPF subscriber headcount, monthly ECR wage challan compliance.
ESIC Portal (Ministry of Labour) Employee State Insurance registration and active contribution status.	ICAI UDIN Registry (ICAI Council) Verification of 18-digit UDIN for CA turnover and net-worth certificates.

CVC-Compliant Cryptographic Audit Trail (/audit-log)

Tamper-evident, SHA-256 chained audit log fulfilling Central Vigilance Commission standards

Immutable SHA-256 Hashing

Non-Repudiation Guarantee

- Every officer action, API response, and document verification generates a cryptographic SHA-256 digest.
- Logs are chained cryptographically: modifying any past record breaks the hash sequence.
- Guarantees that audit evidence cannot be manipulated post-award.

CVC Vigilance Guidelines

Defense Against Arbitration

- Conforms with Central Vigilance Commission (CVC) statutory inspection mandates.
- Records exact timestamps, API latency, auditor IDs, and governing tender clauses.
- Protects procurement officers from frivolous contractor litigation and RTI scrutiny.

1-Click CVC Audit Dossier

Official PDF Export Engine

- Generates an official, publication-quality CVC Compliance Dossier PDF.
- Embeds verifiable QR codes, ELA forensic reports, and portal reconciliation tables.
- Ready for immediate submission to the Competent Financial Authority (CFA).

Tangible Impact for Public Procurement

Demonstrated metrics and national benefits for CPSEs and the GeM ecosystem

99.8%

Time Reduction

Tender scrutiny compressed from 4.5 days down to under 30 seconds.

Rs. 10,000+ Cr

Fraud Shield

Prevents debarred vendors and fake MSME suppliers from claiming preference.

0% Error

Zero Oversight

Automated checks cover 28+ statutory clauses without fatigue or oversight.

100%

Make in India

Protects genuine Class-I domestic manufacturers under national policy.

STRATEGIC ALIGNMENT WITH GOVERNMENT INITIATIVES

- GeM GTC Compliance: Fully implements General Terms & Conditions Clause 4 (Verification of Documentation).
- General Financial Rules (GFR) 2017: Enforces Rule 144(xi) (Land Border Vendor Restrictions) & Rule 151 (Debarment).
- Public Procurement (Preference to Make in India) Order: Strictly protects 50% domestic value-addition threshold.
- Vigilance Non-Repudiation: Meets Central Vigilance Commission (CVC) audit standards with cryptographic integrity.

Modern Enterprise Technology Stack

Built for high availability, zero latency, and military-grade data privacy

Frontend Architecture

- Next.js 14 App Router (React Server Components)
- TypeScript for strict compile-time type safety
- Tailwind CSS with Gov Blue institutional tokens
- Framer Motion for smooth user feedback
- Bilingual Context Engine (English, Hindi, Marathi)

Backend & Forensics Engine

- Next.js API Routes (Serverless / Edge ready)
- PDF-Parse & Canvas Image Layer extraction
- Error Level Analysis (ELA) thermal engine
- PDFKit for CVC Dossier generation
- QRCode engine for real-time mobile validation

Security & Governance

- SHA-256 cryptographic non-repudiation
- Zero-Data Retention for confidential bids
- Role-Based Access Control (Officer / Auditor)
- Sub-50ms gateway query execution
- CVC-compliant cryptographic audit trail

3-Minute Live Presentation Flow for SIH Jury

Step-by-step walkthrough script to impress hackathon evaluators

0:00 - 0:45

Executive Dashboard

Show CPCL Manali scope, point out the 4 KPI cards, and highlight the instant 'Cartel Radar: 1 Ring Detected' banner.

0:45 - 1:30

Bidder Comparison

Navigate to Active Tender. Compare Aura Flow (98% Compliant) with Apex Engineering (8% Disqualified for IOCL Debarment).

1:30 - 2:15

Document Forensics Lab

Open Apex Engineering audit. Toggle ELA Heatmap showing Photoshop turnover alteration (Rs. 1.5 Cr to Rs. 18.5 Cr) and fake UDIN.

2:15 - 3:00

Cartel Radar & CVC Audit

Show Cartel Graph linking Bidder 003 & 004 by Director DIN and Subnet IP. Open Audit Log and download the official CVC PDF Dossier.

THANK YOU

GeM-Rakshak: Safeguarding India's Public Procurement

Summary of Platform Advantages:

- Ultra-Fast Execution: Cuts evaluation time by 99.8% (<30 seconds vs 4.5 days).
- Deep-Tech Forensics: Detects pixel tampering, Photoshop edits, copied QR codes, and fake UDINs.
- 8 Official Gateways: Live querying of GSTN, MCA21, CBDT, Udyam, CPPP, EPFO, ESIC & ICAI.
- Anti-Cartel Radar: Unmasks hidden bidding syndicates and proxy companies sharing directors.
- CVC Compliant: Tamper-evident SHA-256 audit trail with one-click official PDF dossier export.

Live Application: <http://localhost:3000> - GitHub: <https://github.com/SiddhantGpt19/GeM-Rakshak>